

PROBABILITY AND STATISTICS

Lecture-1

Unit-V

Basic Statistics and Correlation

T.Thirupathi

Faculty of Mathematics

Department of Science & Humanities

SNIST

Basic Statistics:

Types of data:-

There are two types of Data

(i) Grouped data

(ii) Un Grouped data

Grouped data :-

If the data is represented in the form of class intervals and their corresponding frequencies then it is said to be a grouped data.

Note:- If the C.I. are in continuous form then the grouped data is said to be closed data otherwise it is open data.

Un Grouped data:-

If the observations are represented randomly without classification then the data is known as ungrouped data.

In this chapter we will going to study the following topics:

(I) Measures of central tendency

(II) Skewness

(III) Kurtosis

(IV) Correlation

(I) Measures of central tendency (or) averages:-

Average represents a whole series as such its value lies between the minimum and maximum values.

Generally it is located in the centre of the distribution.

There are the **five** measures of central tendency that are in common use

(1) Arithmetic Mean(A.M)

or Simple Mean

(2) Median

(3) Mode

(4) Geometric Mean(G.M)

(5) Harmonic Mean(H.M)

(i) (a) Mean or A.M:

For ungrouped data

$x_1, x_2, \dots, x_n$ Mean

Then Mean =

(b) In case of frequency distribution

Eg:

1

0-10	5
10-20	14
20-30	6

$$\text{Mean} = \frac{(5 \times 5) + (15 \times 14) + (25 \times 6)}{5 + 14 + 6} = 15.4$$

Note:

In case of grouped or continuous frequency distribution x is taken as the mid value of the corresponding class.

(c) It may be noted that if the values of x or f are large the calculation of mean by formula is quite time consuming and tedious. The A.M is reduce to a great extent

Where = here A is arbitrary point
and $N = \sum f$, h = class length

(2)Median :-

(a)If a set of observations are arranged in ascending or descending order, then the middle most (or) centre value gives the median.

Note: Median is average of position

Eg:- 1) 4,12,7,9,14,17,16 , median =12

2) 4,12,7,9,14,17,16,21, median = $\frac{12+14}{2} = 13$

Note: 1) If n is odd then $\left(\frac{n+1}{2}\right)^{th}$ term is median.

2) If n is even, then median is the average of $\frac{n}{2}^{th}$ term and $\left(\frac{n}{2} + 1\right)^{th}$ term

(b) In case of discrete frequency distribution median is obtained by considering the cumulative frequencies.

The steps for the calculating median are given below

- (i) Find $N/2$, Where $N =$
- (ii) See the cumulative frequency just greater than $N/2$.
- (iii) The corresponding value of x is median

(c) In case of continuous frequency distribution, the class corresponding to the cumulative distribution just greater than $N/2$ is called the median class. The value of the Median obtained by the following formula:

The value of median is obtained by the following formula:

Median =

Where l = Lower limit of the median class

f = frequency of the median class

h = magnitude of the class

c = the c.f. of the class preceding the median class

(3)Mode:

(a)The observation that occur most frequently
i.e the observation with maximum frequency is called mode.

Ex: (i)The average height of an Indian (male) is

(ii)The average size of the shoes sold in shop is 7

(b) In case of continuous frequency distribution, mode is given by the formula

Mode =

Where L = lower limit of modal class.

=

=

(the highest frequency)

C=Size of the interval

(4) Geometric mean:

(a) Geometric mean of a set of n observations is n th root of their product

Thus the Geometric mean G of n observation

$$GM =$$

(b) In case of continuous frequency distribution

$GM = \text{Antilog}$

Where $N =$

(5) Harmonic Mean:

Harmonic mean of a number of observations is the reciprocal of the arithmetic mean of the reciprocals of the given values.

Thus

(a) H.M of n observations

H.M =

(b) In case of frequency distribution

H.M =

Problem(1).Find (a)Mean (b)
Median (c)Mode (d)Range of
the following data

5,6,2,4,7,8,3,5,6,6

Solution:

(a) Mean

$$=5.2$$

(b)Median:

to find median place all the members in order

2,3,4,5,5,6,6,6,7,8

As there are two middle numbers

5 & 6

median = $\frac{5+6}{2} = 5.5$

(c) Mode: 6 appears more than any other number so mode = 6

(d) Range: It is the difference between the smallest and largest numbers
 $\text{Range} = 8 - 2 = 6$

Problem:2 Calculate Median from the following data:

x	0	1	2	3	4	5	6	7	8
Y	1	9	26	59	72	52	29	7	1

Solution:

x	f	c.f.
0	1	1
1	9	10
2	26	36
3	59	95
4	72	167
5	52	219
6	29	248
7	7	255
8	1	256

Here $N = 256$,
The cumulative frequency just greater than 128 is 167 and the value of x corresponding to 167 is 4.
Hence median is 4.

Problem(3): The frequency distribution of weight in grams of mangoes given below. Calculate the A.M.

Weight in grams	No.of mangoes
410-419	14
420-429	20
430-439	42
440-449	54
450-459	45
460-469	18
470-479	7

Weight in gr	No.of mangoes(f)	Mid-values	d=	fd
409.5-419.5	14	414.5	-3	-42
419.5-429.5	20	424.5	-2	-40
429.5-439.5	42	434.5	-1	-42
L 439.5-449.5	f 54	444.5 A	0	0
449.5-459.5	45	454.5	1	45
459.5-469.5	18	464.5	2	36
469.5-479.5	7	474.5	3	21

To find A.M

We have

$A=444.5$, $= -22$, $N=200$

Mean

$$=444.5 +$$

$$=443.4 \text{ gms}$$

Problem(4): The frequency distribution of weight in grams of mangoes given below. Calculate the Median.

Weight in grams	No.of mangoes
410-419	14
420-429	20
430-439	42
440-449	54
450-459	45
460-469	18
470-479	7

Weight in gr	No.of mangoes(f)	Mid-values	c.f
409.5-419.5	14	414.5	14
419.5-429.5	20	424.5	34
429.5-439.5	42	434.5	C 76
L 439.5-449.5	f 54	444.5 A	130
449.5-459.5	45	454.5	175
459.5-469.5	18	464.5	193
469.5-479.5	7	474.5	200

To find Median

Median =

Where l = Lower limit of the median class

f = frequency of the median class

h = magnitude of the class

c = the c.f. of the class preceding the median class

Here

The c.f is just greater than 100
is 130

Hence the corresponding class
439.5-449.5 is the median
class $l=439.5, h=10, f=54, C=76$

Therefore Median =
 $439.5 + (100 - 76) = 443.94\text{gms}$

Problem(5):
Calculate arithmetic mean and
geometric mean of the following
distribution.

x	2	3	4	5	6	7	8
f	2	4	6	2	3	2	1

x	f	x.f	logx	f.logx
2	2	4	0.3010	0.6020
3	4	12	0.4771	1.9084
4	6	24	0.6021	3.6126
5	2	10	0.6990	1.3980
6	3	18	0.7782	2.3346
7	2	14	0.8451	1.6902
8	1	8	0.9031	0.9031

From table

A.M = =

$$\begin{aligned} \text{GM} &= \text{Antilog} \\ &\text{Antilog} \\ &= 4.192 \end{aligned}$$

Porblem(6). Calculate the mode for the following frequency data.

C.I.	Frequency
10-20	5
20-30	8
30-40	$12(f_{m-1})$
L40-50	$16(f_m)$
50-60	$10(f_{m+1})$
60-70	8

Here $L=40$

$C= 10$

$= 16-12=4$

$= 16-10=6$

Mode =

$=40+10$

$= 44$



**Thanks
for
watching
this video**



PROBABILITY AND STATISTICS

Lecture-2

Unit-V

Basic Statistics and Correlation

T.Thirupathi

Faculty of Mathematics

Department of Science & Humanities

SNIST

Correlation Analysis:

A distribution involving two variables is known as a bivariate distribution.

If these two variables vary such that change in one variable effects the change in other variable , then the variables are said to be correlated .

**The degree of relationship
between the variables
under consideration is
Measured through the
correlation analysis.**

Correlation coefficient or Correlation index:

**The measure of correlation
is called correlation
coefficient**

Types of Correlation:

**(i) Positive correlation and
Negative correlation**

Positive Corelation:

A positive correlation is a relationship between two variables where if one variable increases, the other one also increases. A positive correlation also exists in one decreases and the other also decreases.

Ex(1) The more it rains, the more sales for umbrellas go up.

(2) If more rain and more crop.

(3) As her salary increased, so did her spending.

Negative Coelation:

A negative correlation means that there is an inverse relationship between two variables - when one variable decreases, the other increases. The vice versa is a negative correlation too, in which one variable increases and the other decreases.

Ex(1): A student who has many absences has a decrease in grades.

Ex(2): The more vitamins one takes, the less likely one is to have a deficiency.

Ex(3): If a train increases speed, the length of time to get to the final point decreases.

(ii) Simple and Multiple Correlation:

The *simple Correlation* or *simple linear correlation* between two *variable* say x, y is the relationship that exists between them. *Multiple Correlation*, on the other hand is the *degree of relationship* that exists between three or more variables.

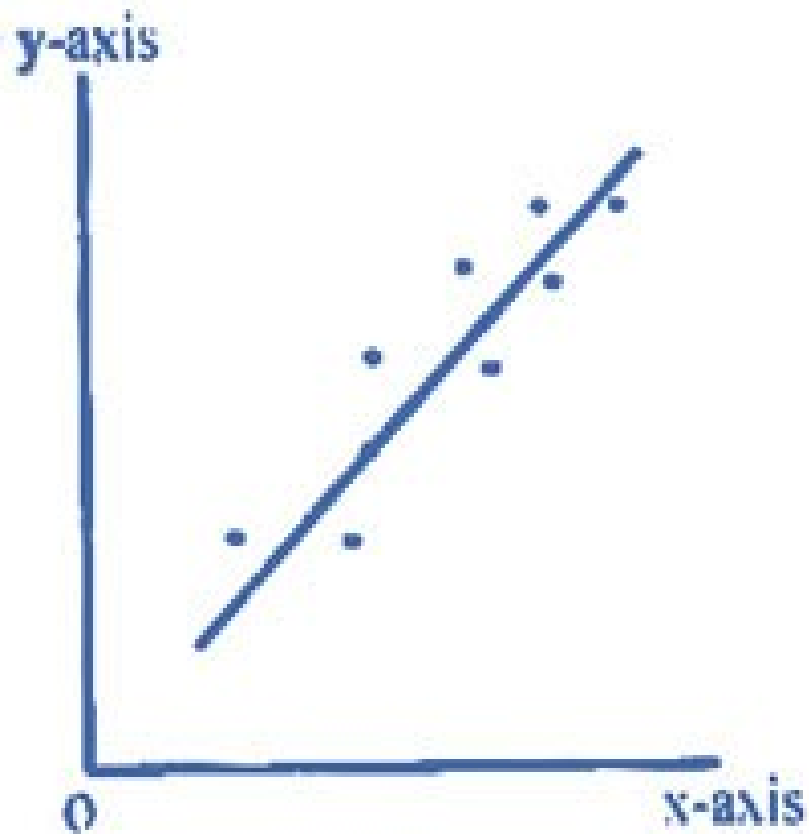
(iii) Linear and Non Linear:

Correlation is said to be linear if the ratio of change is constant.

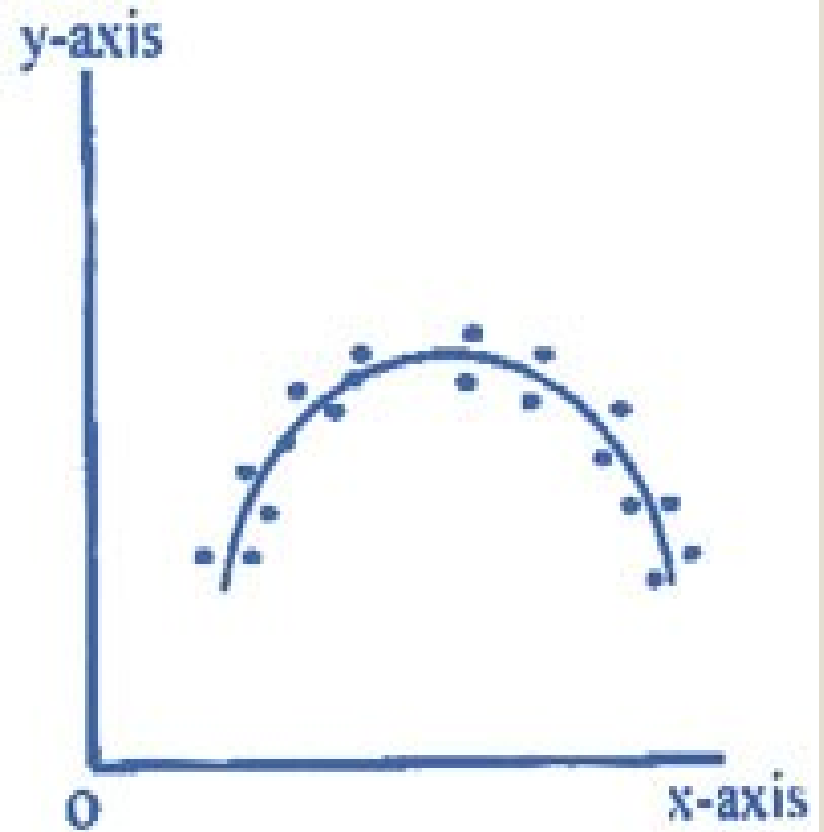
When the amount of output in a factory is doubled by doubling the number of workers, this is an example of linear correlation.

Correlation is said to be non linear if the ratio of change is not constant. In other words, when all the points on the scatter diagram tend to lie near a smooth curve, the correlation is said to be non linear (curvilinear).

This is shown in the figure on the right below.



Linear Correlation



Non Linear Correlation

Methods of Studying Correlation:

(I) Scatter Diagram method

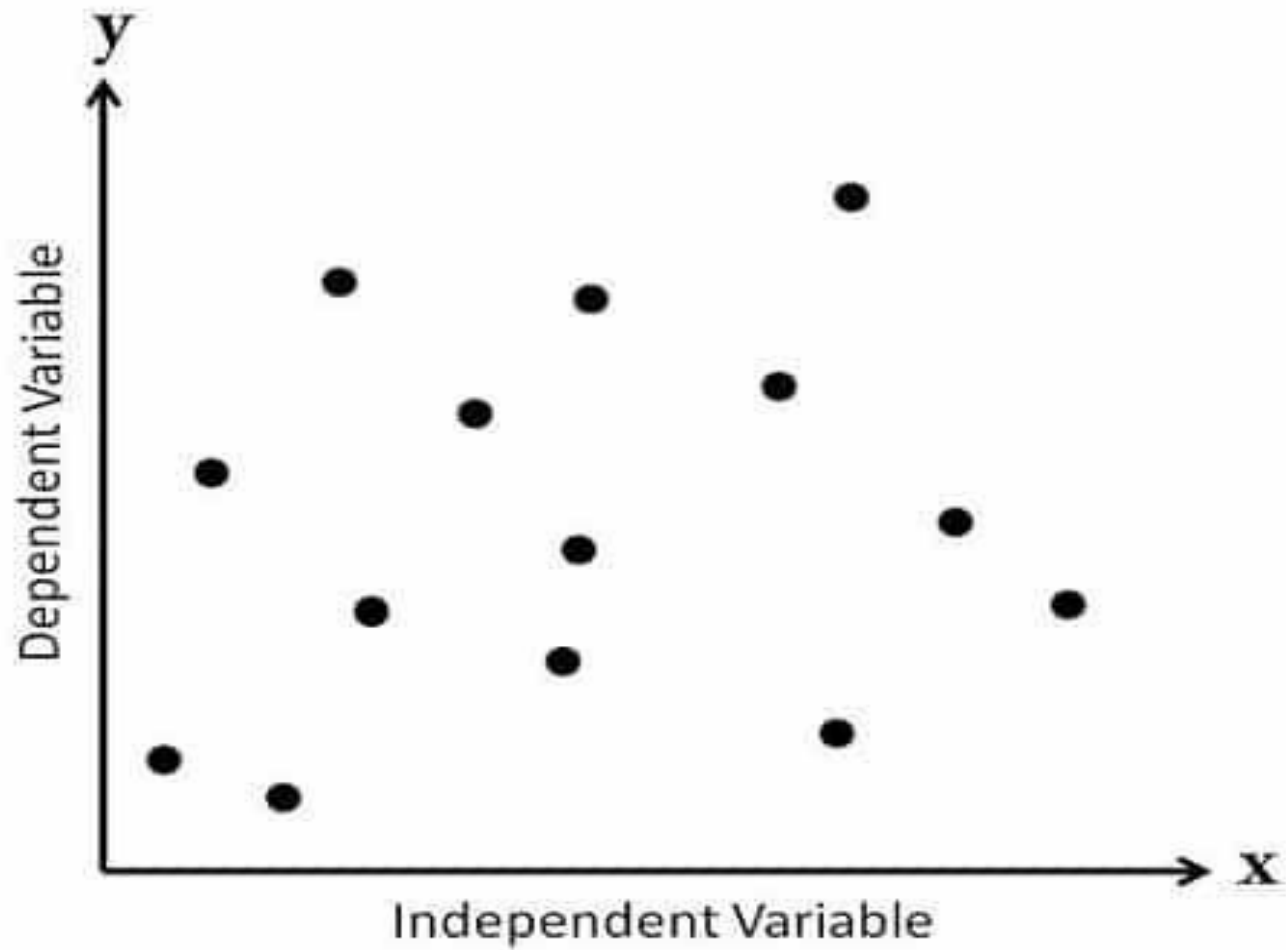
**(II) Karl Pearson's
Coefficient of Correlation**

**(III) Rank method or
Spearman's Rank Coefficient
Correlation**

(I) Scatter Diagram method:

A scatter plot, scatter graph, and correlation chart are other names for a scatter diagram.

We draw this graph with two variables. The first variable is independent and the second variable depends on the first.



Types of Scatter Diagram:

According to the correlation, you can divide

scatter diagrams into the following categories:

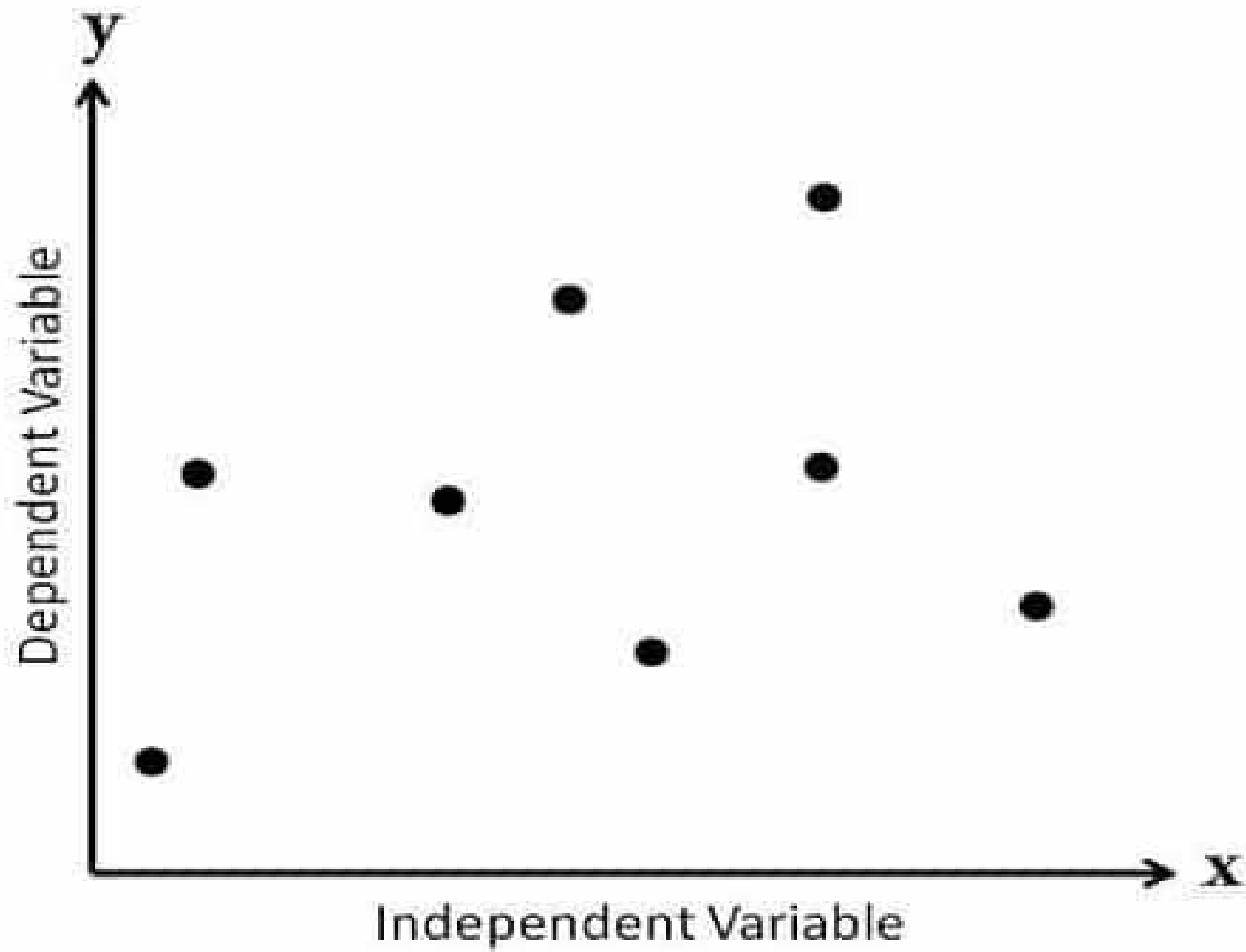
(i) Scatter Diagram with No Correlation

(ii) Scatter Diagram with Moderate Correlation

(iii) Scatter Diagram with Strong

(i) Scatter Diagram with No Correlation:

This diagram is also known as “Scatter Diagram with Zero Degree of Correlation”.



Here, the data point spread is so random that you cannot draw a line through them.

Therefore, you can say that these variables have no correlation.

(ii) Scatter Diagram with Moderate Correlation:

This diagram is also known as "Scatter Diagram with a Low Degree of Correlation".



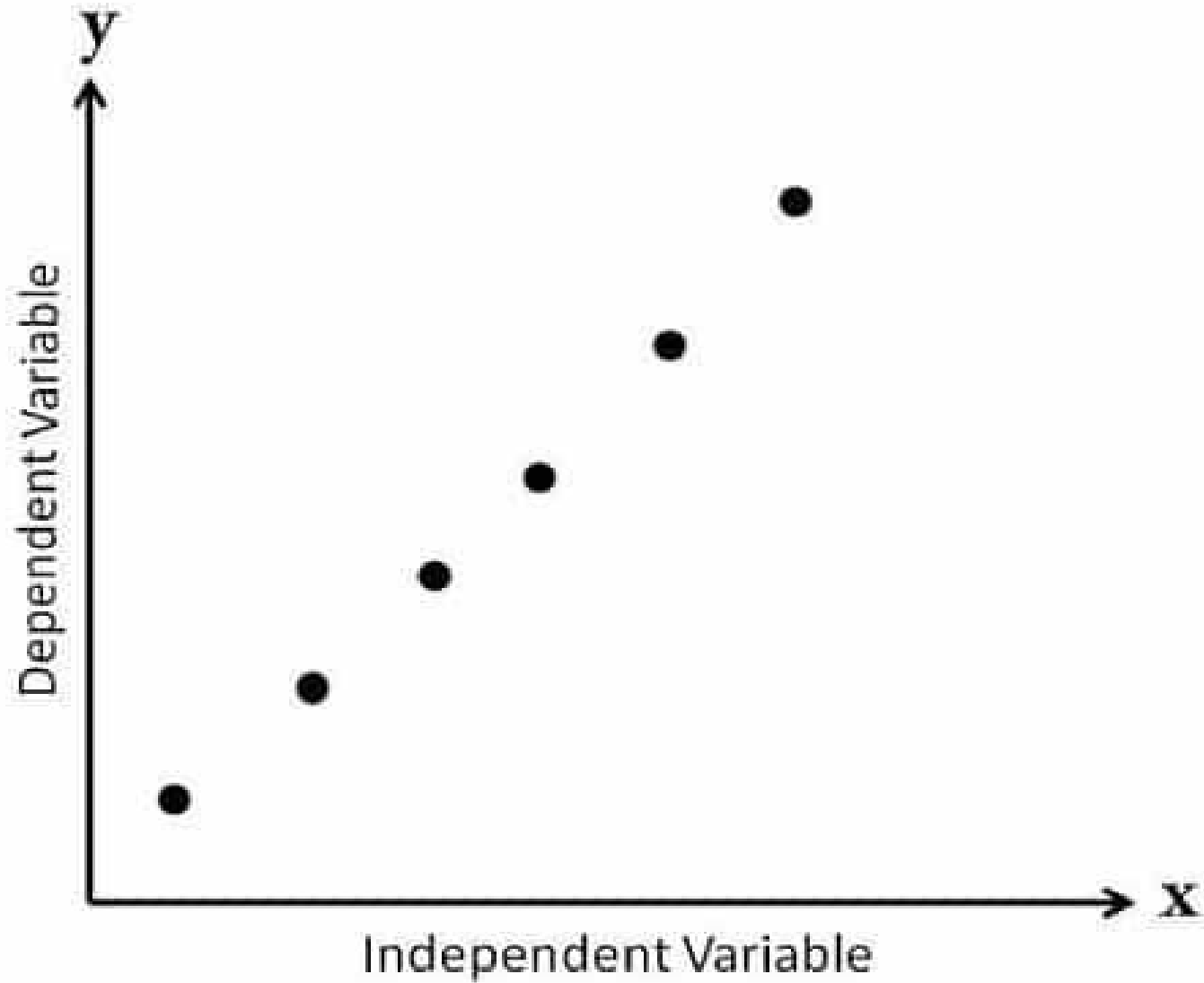
This diagram is also known as “Scatter Diagram with a Low Degree of Correlation”.

Here, the data points are a little closer and you can see that some kind of relationship exists between these variables.

(iii) Scatter Diagram with Strong Correlation

This diagram is also known as “Scatter Diagram with a High Degree of Correlation”.

In this diagram, data points are close to each other and you can draw a line by following their pattern.



In this case, you say that these variables are closely related.



**Thanks
for
watching
this video**



PROBABILITY AND STATISTICS

Lecture-4

Unit-V

Basic Statistics and Correlation

T.Thirupathi

Faculty of Mathematics

Department of Science & Humanities

SNIST

(II) Karl Pearson's Coefficient of Correlation :

Karl Pearson a British statistician suggested mathematical method for measuring the magnitude of linear relationship between two variables.

This is Known as Pearson's
Coefficient of Correlation
It is denoted by " r "

There are 2 formulae for r

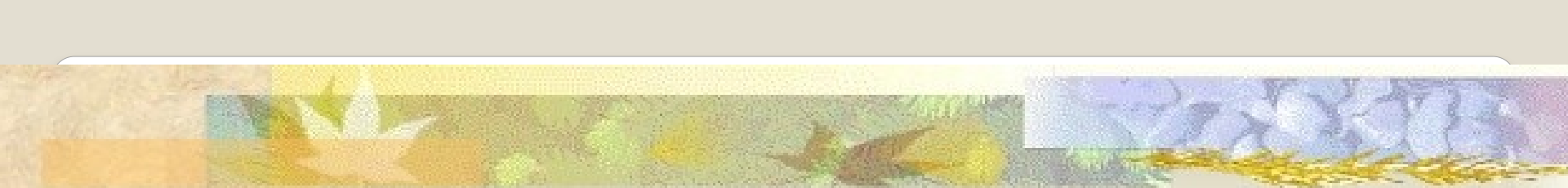
Formula(i): Direct method

$r =$

Where $=$
 $=$

$=$ Mean of series $=$

$=$ Mean of series $=$



Interpretation of Correlation Coefficient (r)

- The value of correlation coefficient ' r ' ranges from -1 to +1
- If $r = +1$, then the correlation between the two variables is said to be perfect and positive
- If $r = -1$, then the correlation between the two variables is said to be perfect and negative
- If $r = 0$, then there exists no correlation between the variables

Problem(1): Calculate Karl Pearson's coefficient of correlation from the following data.

Wages ()	100	101	102	102	100	99	97	98	96	95
Cost of Living()	98	99	99	97	95	92	95	94	90	91

Solution:

**We have Karl pearson
coefficient of correlation is**

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (1)$$

Where $\bar{x} =$

$\bar{y} =$

$\bar{x} =$ Mean of series $\bar{y} =$

$\bar{y} =$ Mean of series $\bar{x} =$

Here =

=

= 99

=

=

= 95

		-	- =			
100	98	1	3	3	1	9
101	99	2	4	8	4	16
102	99	3	4	12	9	16
102	97	3	2	6	9	4
100	95	1	0	0	1	0
99	92	0	-3	0	0	9
97	95	-2	0	0	4	0
98	94	-1	-1	1	1	1
96	90	-3	-5	15	9	25
95	91	-4	-4	16	16	16
				=61	=54	=96

Substituting table values

=61 =54 =96

in equation(1)

i.e

r=

we get

r=

Problem(2): Psychological tests of intelligence and of engineering ability were applied to 10 students. Here a record of data showing intelligence ratio(I.R) and Engineering ratio (E.R) given. Calculate Karl Pearson's coefficient of correlation.

Stud ent	A	B	C	D	E	F	G	H	I	J
I.R	105	104	102	101	100	99	99	98	93	92
E.R	101	103	100	98	95	96	104	92	97	94

Solution:

We have Karl pearson coefficient of correlation is

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (1)$$

Where $\bar{x} =$
 $\bar{y} =$

$\bar{x} =$ Mean of series $\bar{y} =$

$\bar{y} =$ Mean of series $\bar{x} =$

Here =

=

= 99

=

=

= 98

		- -99=	-= -98=			
105	101	6	3	18	36	9
104	103	5	5	25	25	25
102	100	3	2	6	9	4
101	98	2	0	0	4	0
100	95	1	-3	-3	1	9
99	96	0	-2	0	0	4
98	104	-1	6	-6	1	36
96	92	-3	-6	18	9	36
93	97	-6	-1	6	36	1
92	94	-7	-4	28	49	16
				=92	=170	=140

From table we have

$$=92 \quad =170 \quad =140$$

**Substitute these values in
equation(1),**

i.e $r =$

We get

$$r = 0.59$$

Problem(3): Calculate coefficient of correlation between age of cars and annual maintenance cost and comment

(Ans: $r=0.836$)

Age of cars(y ears)	2	4	6	7	8	10	12
Annual Maintenance (Rupees)	1600	1500	1800	1900	1700	2100	2000

Problem(4): Calculate Karl Pearson's coefficient of correlation from the following data
(Ans: $r=0.65$)

Potato per kg	18	18	18	18	20	20	21
Tomoto per kg	26	31	28	28	31	35	31

Formula(ii)

**When deviation taken from
assumed mean:**

**When actual mean is not a
whole number but a fraction
or when the series is large**

**the calculation by direct
method will involve a lot
of time to avoid such
Calculation We can use
assumed mean method**

In this case

$r =$

Where

A,B are assumed mean

Problem(1): Calculate Karl Pearson's coefficient of correlation from the following data.

	28	41	40	38	35	33	40	32	36	33
	23	34	33	34	30	26	28	31	36	38

Solution:

$$r = \frac{\bar{y} - \mu}{\sigma} \quad (1)$$

Where

$\bar{y}$ = assumed means

Here =

=

= 35.6

So it is a decimal

Therefore assume that

=

=

= 31.3

So it is a decimal

Therefore assume that B

x	y	X=	Y=	XY		
28	23	-7	-8	56	49	64
41	34	6	3	18	36	9
40	33	5	2	10	25	4
38	34	3	3	9	9	9
35	30	0	-1	0	0	1
33	26	-2	-5	10	4	25
40	28	5	-3	-15	25	9
32	31	-3	0	0	9	0
36	36	1	5	5	1	25
33	38	-2	7	-14	4	49

**Substituting all these table
values in equation (1)**

We get

$$r =$$

$$= 0.4402$$

Problem(2):Calculate Karl Pearson's coefficient of correlation from the following data.
(Answer: $r=0.472$)

Height of father (inches)	65	66	67	67	68	69	71	73
Height of son (inches)	67	68	64	68	72	70	69	70



**Thanks
for
watching
this video**



PROBABILITY AND STATISTICS

Lecture-5

Unit-V

Basic Statistics and Correlation

T.Thirupathi

Faculty of Mathematics

Department of Science & Humanities

SNIST

Rank Correlation coefficient:

A British psychologist Spearman was given the method of find the coefficient by ranks. This method is based on rank and is useful in dealing with qualitative characteristics

such as morality, character,
intelligence and beauty.

It is based on ranks given to
the observations.

The formula for Spearman
Rank Correlation coefficient

=

= **Spearman Rank
Correlation coefficient**

= Sum of the squares of
difference of 2 ranks

n = number of paired
observation

Properties:

(i) The value of τ lies between -1 and +1

(ii) If $\tau = +1$, there is a complete agreement in the order of ranks and move in the same direction.

(iii) If $= -1$, there is a complete disagreement in the order of ranks and they are in opposite directions.

(iv) $= 0$, there is no association in the ranks.

Problem(1):Following are the ranks obtained by 10 students in 2 subjects Mathematics and Statistics. Find the rank Correlation coefficient.

Statistic s	1	2	3	4	5	6	7	8	9	10
Math emat ics	2	4	1	5	3	9	7	10	6	8

Solution:

The formula for Spearman Rank
Correlation coefficient

$$= 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \quad (1)$$

= **Spearman Rank**

Correlation coefficient

= Sum of the squares of difference
of 2 ranks

n = number of paired observations

Statistics(x)	Mathematics(y)	D=x-y	
1	2	-1	1
2	4	-2	4
3	1	2	4
4	5	-1	1
5	3	2	4
6	9	-3	9
7	7	0	0
8	10	-2	4
9	6	3	9
10	8	2	4

From table

We have $n=10$

Substitute in equation (1),
we get

$=$

$= 0.7576$

Problem(2): Ten competitors in a musical test were ranked by the three judges A,B and C in the following order.

Ranks by A	1	6	5	10	3	2	4	9	7	8
Ranks by B	3	5	8	4	7	10	2	1	6	9
Ranks by C	6	4	9	8	1	2	3	10	5	7

Using rank correlation method,
discuss which pair of judges has
the nearest approach to common
liking in music.

Solution:

$$= \text{-----}(1)$$

= Spearman Rank Correlation
coefficient

=Sum of the squares of
difference of 2 ranks

n=number of paired
observation

Ranks by A (X)	Ranks by B (Y)	Ranks by C (Z)						
$\sum D_L = 0$								
1	3	6	-2	-5	-3	4	25	9
6	5	4	1	2	1	1	4	1
10	4	8	6	2	-4	36	4	16
3	7	1	-4	2	6	16	4	36
2	10	2	-8	0	8	64	0	64
4	2	3	2	4	-1	4	1	1
9	1	10	8	-1	-9	64	1	81
7	6	5	1	2	1	1	4	1
8	9	7	-1	1	2	1	1	4
Total							=60	=214

$$\rho_1(X, Y) = 1 - \frac{6 \sum D_1^2}{n(n^2 - 1)} = 1 - \frac{6 \times 200}{10 \times 99} = 1 - \frac{40}{33} = \frac{-7}{33}$$

$$\rho_2(X, Z) = 1 - \frac{6 \sum D_2^2}{n(n^2 - 1)} = 1 - \frac{6 \times 60}{10 \times 99} = 1 - \frac{4}{11} = \frac{7}{11}$$

$$\rho_3(Y, Z) = 1 - \frac{6 \sum D_3^2}{n(n^2 - 1)} = 1 - \frac{6 \times 214}{10 \times 99} = 1 - \frac{214}{165} = \frac{-49}{165}$$

Since (X,Z) is maximum,
we conclude that the pair of
judges A and C has the
nearest approach to common
likings in music.

Equal or Repeated Rank:

If there are more items with same values then common ranks are given to repeated items . This common rank is average of marks

Ex: If 2 individuals have been placed in place then the rank is $=7.5$

If 3 individuals have been placed in place then the rank is $=8$

The Spearman Rank Correlation coefficient for repeated ranks is given by

$$\rho = 1 - 6 \left[\frac{\sum D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \dots}{n(n^2 - 1)} \right]$$

Where m is no. of items whose rank is common.

Example 1: From the following data calculate the rank correlation coefficient after making adjustment for tied ranks.

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

X	Ranks (X)	Y	Ranks (Y)	$D = X - Y$	D^2
48	8	13	5.5	2.5	6.25
33	6	13	5.5	0.5	0.25
40	7	24	10	-3	9.00
9	1	6	2.5	-1.5	2.25
16	3	15	7	4	16.00
16	3	4	1	2	4.00
65	10	20	9	1	1.00
24	5	9	4	1	1.00
16	3	6	2.5	5	0.25
57	9	19	8	1	1.00
					$\sum D^2 = 41$

16 is repeated 3 times in X items
hence $m=3$.

Since 13 is repeated twice in Y
items hence $m=2$.

Since 6 is repeated twice in Y item
hence $m=2$.

$$\therefore \rho = 1 - \frac{6 \left[\sum D^2 + \frac{1}{12}(m^2 - m) + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) \right]}{n^3 - n}$$

$$= 1 - \frac{6 \left[41 + \frac{1}{12}(3^3 - 3) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right]}{990} = +0.733$$

Problem(2): Obtain the rank correlation coefficient for the following data.

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

X	Y	Ranks (X)	Ranks (Y)	$D = X - Y$	D^2
68	62	4	5	-1	1
64	58	6	7	-1	1
75	68	2.5	3.5	-1	1
50	45	9	10	-1	1
64	81	6	1	5	25
80	60	1	6	-5	25
75	68	2.5	3.5	-1	1
40	48	10	9	1	1
55	50	8	8	0	0
64	70	6	2	4	16
				$\sum D = 0$	$\sum D^2 = 72$

75 is repeated 2 times in X items hence $m=2$.

Since 64 is repeated 3 times in X series hence $m=3$.

Since 68 is repeated 2 times in Y series hence $m=2$.

$$\therefore \rho = 1 - \frac{6 \left[\sum D^2 + \frac{1}{12} (m^2 - m) + \frac{1}{12} (m^3 - m) + \frac{1}{12} (m^3 - m) \right]}{n^3 - n}$$

$$= 1 - \frac{6 \left[72 + \frac{1}{12}(3^3 - 3) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right]}{990}$$
$$= 0.4545$$



**Thanks
for
watching
this video**

